QG-COVER:
Quality-Gated Coverage for 3D Gaussian Splatting from Casual Captures

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Abstract

Casual, handheld capture for novel view synthesis yields image pools that are both highly redundant and quality-varying, containing underexposed, noisy, or motion/defocus-blurred frames. We study view selection for 3D Gaussian Splatting (3DGS) under a fixed training-view budget and show that coverage-only selection is unreliable in this regime: Farthest-View Sampling (FVS) favors geometrically extreme viewpoints, which in casual capture coincide with degraded frames, so it can even underperform random selection. A training-free no-reference quality gate placed in front of coverage selection recovers $+\mathbf{1.69}$ *dB over FVS on seven mip-NeRF-360 scenes with* 30% *corrupted frames at a* 40%*-of-pool budget;* Quality-Gated Coverage *(*QG-COVER*), a quality-weighted submodular facility-location objective that covers every viewpoint while favoring high-quality source views, raises the gain to* $+\mathbf{1.95}$ *dB, leads at every budget tier, and degrades gracefully when clean frames are scarce. That same subset even beats training on the whole corrupted pool, yet still lands* 1.57 *dB below clean-pool training: selection is damage control, not free information. On* 26 *real casual captures (nerfbusters, DL3DV)* QG-COVER *beats FVS on every scene at the tightest budget tier, and our gate independently flags a comparable fraction of their frames as low quality—this regime arises naturally in casual capture rather than by our construction. Retraining all* 33 *scenes at the standard* $30k$ *3DGS schedule preserves the matched-budget advantage, while the weaker subset-beats-pool result is schedule- and domain-sensitive; we report where it breaks.*

1. Introduction

3D Gaussian Splatting [6] has made photorealistic, real-time novel-view synthesis broadly accessible, and the images that drive it are increasingly captured *casually*—a person walking around an object or a room with a phone. Such capture is unavoidably *redundant* (video at 30 fps yields hundreds of near-duplicate frames) and, less often acknowledged, *quality-varying*: hand motion blurs frames, exposure swings, autofocus hunts, and low light adds sensor noise—within a single real scene, sharpness varies by more than an order of magnitude (supplement).

Redundancy invites *view selection*: rather than train on every frame (slow, and as we show, harmful *once the pool is quality-varying*), pick an informative budget-K subset. That qualifier is the entire scope of this paper: on a clean pool, training on everything is optimal on all seven uncorrupted mip-NeRF-360 scenes. What changes in casual capture is that a blurred or underexposed frame is a *wrong* constraint, not a weak one, while redundancy makes its coverage available elsewhere. Fig. 1 previews both halves: a clean pool improves monotonically with the budget while a corrupted one peaks below the full pool, and it is the standard coverage criterion that keeps the frames a cheap quality gate rejects. That asymmetry—not the selection algorithm itself—is why a well-chosen 40% can beat the whole corrupted pool; Sec. 4 quantifies both effects. Prior view-acquisition signals—information gain [5, 21], geometric coverage [33], render quality [30]—all assume every candidate is a valid image, so the only question is which *viewpoints* to add (Sec. 2); we show that assumption is load-bearing.

Consider a phone video orbiting a plant. Farthest-View Sampling [33], the canonical coverage baseline, seeks the K maximally spread camera centers—but the geometrically extreme viewpoints are captured near the orbit's direction changes, where capture instability, exposure variation, and focus excursions are common. FVS therefore systematically over-selects the blurriest frames, and 3DGS trained on that subset floats and smears texture—not merely suboptimal but counterproductive: FVS falls *below* uniform random selection on up to 13 of our 33 scenes.

The fix is cheap: a training-free no-reference gate built

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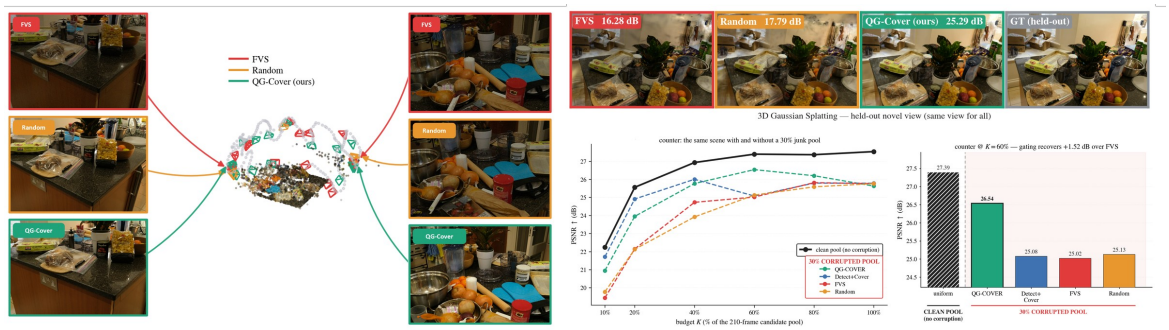


Figure 1. **QG-COVER: quality-gated coverage for casual-capture 3DGS** (counter, 30% low-light corruption; all panels are measured). *Left:* the capture's cameras and point cloud with each selector's picks: coverage-only FVS (red) keeps $14/40$ corrupted views while our gate (green) rejects them all ($0/40$); random (amber) is indiscriminate; call-outs show the kept frames at two contested viewpoints. *Top right:* on a held-out view FVS renders poorly (16.3 dB) and QG-COVER sharply (25.3 dB). *Bottom right:* sweeping the budget from 10 to 100% of the pool separates the two regimes—on the *clean* pool (solid black) more views is monotonically better, while on the corrupted pool every selector peaks below 100% and gated coverage dominates FVS at every budget; the bar chart cuts the sweep at $K{=}60\%$, where gating recovers $+1.52$ dB over FVS yet still sits 0.85 dB below the clean-pool reference.

from three classical image statistics (Sec. 3.2) detects corrupted frames reliably, erring toward rejecting clean frames rather than admitting junk, recovers most of the loss, and comes close to an *oracle* gate told which frames are corrupted (supplement).

A hard gate, however, is brittle when clean frames are scarce. We therefore recast selection as a single quality-weighted submodular objective, QG-COVER (Sec. 3.4)—cover the entire viewpoint distribution while *favoring* high-quality source views—which matches the hard gate in the easy regime and stays above FVS at the corruption level where the hard gate drops below it.

Synthetic corruption yields the per-view ground-truth labels needed for the oracle and detector; to rule out a corruption-model artifact we validate on real captures with natural quality variation (nerfbusters [31], DL3DV [13]), where the effect is smaller but consistent: at the tightest budget tier QG-COVER improves over FVS on every real scene ($26/26$). Selecting on quality alone fails symmetrically: a sharpness-maximizing selector falls below FVS on *every* real scene, because the sharpest frames cluster temporally and destroy coverage—quality *and* coverage are both necessary.

In summary, we contribute:

- A counterintuitive **diagnosis**: geometric coverage-maximization is unreliable on a quality-varying pool—it wins on some scenes and loses to random selection on others, with no way to tell which in advance (FVS falls below random on 8–13 of 33 scenes depending on budget)—and a foundation-embedding (DINOv2) selector adds nothing once the pool is gated. We trace this to a mechanism: coverage-maximizing selection over-selects corrupted boundary views (on real captures they cluster near trajectory reversals), bounding the "coverage beats information-gain" conclusion of prior work to a specific regime.
- An **empirical link** from no-reference image quality to 3DGS quality: our training-free gate converts the several-dB harm of corrupted views into a downstream gain, winning on all 26 real scenes at the tightest budget.
- **QG-COVER**: a quality-weighted submodular facility-location selector that separates *where* to cover from *which view* covers it—so a corrupted view is covered by a nearby clean one instead of by itself—with a $(1 - 1/e)$ greedy guarantee on the coverage objective and graceful degradation under heavy corruption.

2. Related Work

2.1. View and frame selection for neural rendering

Training-view and next-best-view selection uses three acquisition signals, typically evaluated on clean or curated candidate pools. *Information / uncertainty* methods acquire views that most reduce model uncertainty [5, 21, 34, 35]. *Geometric coverage* methods spread viewpoints: NeRF-Director [33] studies Farthest-View Sampling and argues, *on clean data*, that coverage can beat uncertainty-guided selection—the conclusion our corrupted-pool regime inverts. *Render-quality* methods score the model's output: Active View Selector (AVS) [30] predicts a candidate's rendered SSIM and acquires the worst-rendering viewpoint. These methods perform *active acquisition* of new views, whereas ours is offline *pruning* of a fixed pool; AVS is closest in spirit but *opposed* in signal—scoring *rendered outputs* from a clean pool, it reads a corrupted input, which renders poorly, as high-priority and *selects* it: exactly the frames we reject from the raw *inputs*.

2.2. Robust and degradation-aware 3DGS/NeRF

A parallel literature makes reconstruction tolerant to corrupted inputs by *repairing in training* over all views: robust losses down-weight transient distractor pixels [25–27], per-image embeddings absorb photometric variation [15], and per-view forward models target a *single* corruption (defocus/motion blur [10, 36], illumination [2]); a recent method [4] soft-masks unreliable Gaussians. Richer scene models handle other facets of casual capture: Nerfies [22] fits a deformation field with per-image appearance codes, WildGaussians [9] combines appearance embeddings with DINO-based uncertainty to suppress occluders and illumination changes, and HQGS [12] adds structural guidance to reconstruct from degraded inputs. These keep every view, need a corruption-specific model or a semantic outlier region, and are silent on capture budget. We differ in three ways: our corruption is *whole-frame* (low-light, noise, defocus), not the sparse per-pixel outliers these losses mask; we *reject* corrupted views rather than modeling them, trainer-agnostically and near-free; and in a redundant pool rejecting a viewpoint costs little coverage, so a gate is a complementary front-end—gate first, then restore the survivors with a robust trainer if desired. Such repair-based trainers [2, 10, 12, 36] ask how to train on a corrupted view, whereas we ask which views to train on; the two compose, and we leave their combination to future work.

2.3. No-reference image quality assessment

NR-IQA spans natural-scene-statistics models [17, 18], focus/sharpness operators [23], and learned predictors. In 3D pipelines, blur filtering is commonly applied *before* reconstruction, e.g. to prune frames for Structure-from-Motion (SfM)/photogrammetry [3, 29], where it is typically evaluated by runtime savings or blur labels rather than by downstream 3D fidelity. To our knowledge this link has not been measured directly for 3DGS; we do so, evaluating the calibrated gate as a *detector* against per-view ground truth and by its downstream gain. We claim novelty not for any single filter but for their robust combination and its measured effect on 3DGS quality.

2.4. Submodular and coreset subset selection

Joint quality/diversity selection is well formalized: the $(1-1/e)$ greedy guarantee [19], DPP quality-diversity kernels [8], facility-location/k-center core-sets [28], filter-then-cover active learning [32], and coresets for efficient training [7, 16]. These target classification and summarization, where a "bad" item is an outlier to prune; our corrupted view is instead *in-distribution content at a valuable pose*, and QG-COVER keeps the viewpoint as a coverage target while restricting the covering *source* to gated-clean views—a decoupling absent from DPP/coreset formulations.

3. Method

Our pipeline has two stages—a quality gate and a coverage selector—which we present first separately (DETECT+COVER) and then fuse into a single objective (QG-COVER); Fig. 2 gives an overview. All components are training-free and depend on no learned model.

3.1. Problem setup and preliminaries

3D Gaussian Splatting. 3DGS [6] optimizes view-dependent 3D Gaussians from posed images by differentiable rasterization under a photometric loss; reconstruction quality depends on both the fidelity of the training views and their coverage of the viewpoint manifold.

Casual-capture pool. We are given a pool $V = \{(I_i, c_i)\}_{i=1}^{N}$ of posed frames, where I_i is an image and $c_i \in \mathbb{R}^3$ its camera center (recovered by COLMAP for real captures). An *unknown* subset $C \subset V$ is corrupted (low-light, noise, or defocus/motion blur). We must choose a subset $S \subseteq V$ with $|S| = K$ to train a 3DGS model $\Theta(S)$, and we measure held-out novel-view synthesis quality on a set of test views disjoint from V. A fixed budget is the standard protocol of the view-selection literature [5, 33] and separates *which* views are chosen from *how many*; in practice it arises when a frame cap, storage, or per-view preprocessing bounds what a pipeline can ingest. We assume nothing about C at selection time; pose recovery for unposed casual video [11] is orthogonal.

3.2. A training-free no-reference quality gate

For each view we compute three classical, training-free features. *(i) Luminance* L_i is the mean of the grayscale image; low-light frames have low L. *(ii) Sharpness* S_i is the Tenengrad gradient energy (mean squared Sobel gradient magnitude); defocus and motion blur lower S. *(iii) Noise* N_i is the standard deviation of the high-frequency residual $g - \mathrm{box}_3(g)$ (the image minus its 3×3 box-blur); sensor noise raises N.

A single feature is a brittle detector: sensor noise *inflates* a raw sharpness measure (noise looks like high-frequency detail), while a blurred frame is smooth and low-noise, so each single-feature threshold fails on some off-target corruption (the supplement quantifies both failures). We therefore convert each feature into *one-sided corruption evidence* by robust z-scoring within the scene (median/MAD, resistant to the corrupted minority) and a rectifier:

$$\begin{aligned} \mathrm{corr}_i \;&=\; \mathrm{ReLU}\big(-z(L_i)\big) + \mathrm{ReLU}\big(-z(S_i)\big) \\ &\quad+\; \mathrm{ReLU}\big(z(N_i)\big), \qquad q_i \;=\; -\,\mathrm{corr}_i. \end{aligned} \tag{1}$$

Intuitively, a frame accrues corruption evidence for being darker, less sharp, *or* noisier than its scene's typical frame,

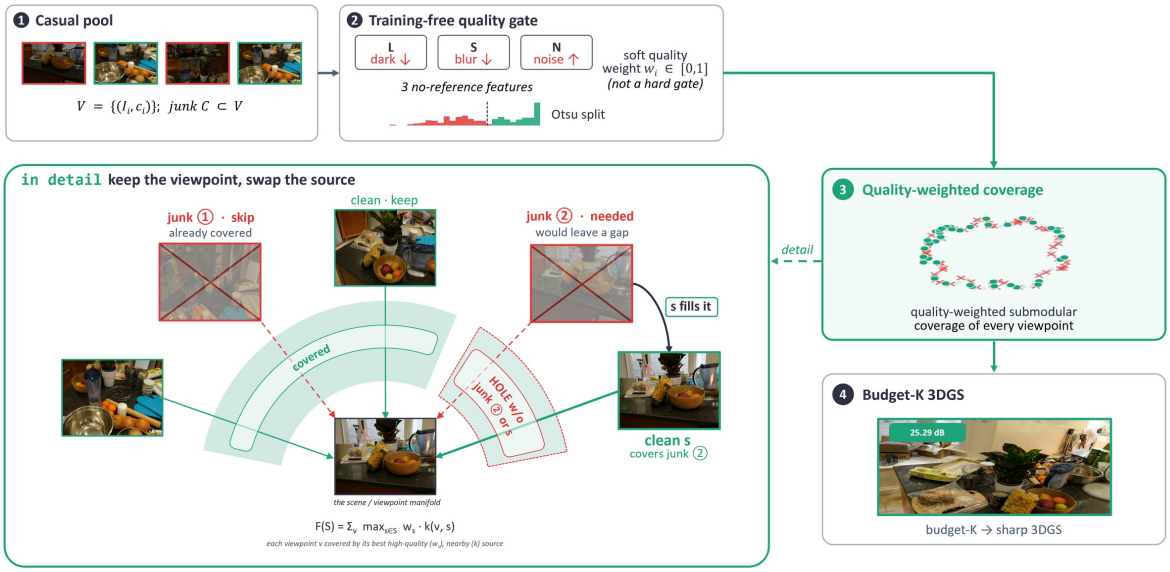


Figure 2. **Overview of QG-COVER**, illustrated end-to-end on counter (30% low-light corruption). **(1)** A casual pool V mixes clean and junk frames. **(2)** A training-free gate scores each view from luminance L, Tenengrad sharpness S and noise N, mapped to a soft weight $w_i \in [0, 1]$ by 5–95th-percentile normalization (Otsu's threshold builds the hard gate of DETECT+COVER). **(3)** The objective $F(S) = \sum_v \max_{s \in S} w_s\, k(v, s)$ covers every viewpoint while favoring high-quality source views: a redundant junk viewpoint (①) is skipped because a clean neighbor already overlaps it, whereas a coverage-critical one (②) would leave a hole, so the nearest clean view s fills it—"keep the viewpoint, swap the source". **(4)** The budget-K subset trains a sharp 3DGS model.

and no benign feature can cancel evidence from another. A view is gated *clean* if its quality score q_i exceeds an Otsu threshold on $\{q_i\}$, providing a scene-adaptive split without requiring the corruption fraction a priori. While the individual features are well-established, our contribution is their robust one-sided *combination* and its use as a calibrated front-end for view selection. *Implementation.* All features are computed on the grayscale image (Sobel gradients for S, a 3×3 box filter for N), $z(\cdot)$ uses the per-scene median and MAD, and the Otsu threshold is taken on the scene's $\{q_i\}$; no constant is tuned per scene or dataset.

3.3. Failure analysis of coverage-only selection

Farthest-View Sampling (FVS) is farthest-point sampling over camera centers, choosing the K most spread-out views. On a casual orbit or sweep the geometric extremes occur near trajectory reversals and sweep boundaries, where handheld motion, exposure, and focus become unstable, so they are *disproportionately* likely to be degraded. FVS thus concentrates its budget in exactly the frames a quality gate would reject (on nerfbusters/century it places 18 of 40 picks in the blurriest 30%, versus 8 for QG-COVER; supplement). Because 3DGS has no per-frame robustness, these corrupted anchors inject floaters and blur that the photometric loss spreads across the reconstruction. The failure is systematic—a coupling between the geometric extremity FVS rewards and the corruption it ignores. This over-selection mechanism is specific to *real* capture, where corruption clusters at motion extremes: in our *synthetic* controls corruption follows an Independent and Identically Distributed (i.i.d.) process and is thus near-independent of camera geometry, so even FVS selects junk close to the pool base rate (as does the information-gain FisherRF selector, $\approx$ 29% vs a 30% pool; supplement). FVS still lands slightly above that base rate ($14/40$ on counter, Fig. 1), but this is too small to explain why it falls *below* random indoors; we attribute that deficit to a coverage/test-distribution mismatch—its picks concentrate on trajectory extremes—rather than to over-selection of corrupted frames, which the real-data selection statistics establish on their own.

The simplest deployable fix runs FVS over the camera centers of the gated-clean views (falling back to the full pool if fewer than K survive); this DETECT+COVER baseline removes essentially all corrupted frames and captures the bulk of the gain (Sec. 4).

3.4. QG-COVER: quality-weighted submodular coverage

A hard gate discards views and is brittle when clean frames are scarce. We instead solve a single objective that fuses quality and coverage. Let $w_i = \mathrm{clip}\big((q_i - P_5)/(P_{95} - P_5),\, 0,\, 1\big) \in [0, 1]$ be the soft-normalized quality of view i, where P_5, P_{95} are the 5th/95th percentiles of $\{q_i\}$ over

Algorithm 1 QG-COVER greedy selection

Input: centers $\{c_i\}$, weights $\{w_i\}$, budget K
$\sigma \leftarrow$ median-NN$(\{c_i\})$; $A_{vs} \leftarrow w_s \exp(-\|c_v - c_s\|^2/2\sigma^2)$ $\forall v, s$ ▷ $N \times N$, cached once
$S \leftarrow \emptyset$; $b_v \leftarrow 0$ $\forall v$ ▷ best coverage of each pool view so far
for $t = 1 \dots K$ **do**
  $s^\star \leftarrow \arg\max_{s \notin S} \sum_v \big(\max(b_v, A_{vs}) - b_v\big)$
  $S \leftarrow S \cup \{s^\star\}$; $b_v \leftarrow \max(b_v, A_{vs^\star})$ $\forall v$
end for
return S

the pool, and let $k(v,s) = \exp\big(-\|c_v - c_s\|^2/2\sigma^2\big)$ be a translation-only viewpoint-similarity kernel whose bandwidth σ is the median nearest-neighbor camera spacing. We maximize the quality-weighted facility-location coverage of the *whole* pool V using only the selected views S:

$$F(S) = \sum_{v \in V} \max_{s \in S} w_s \, k(v,s). \tag{2}$$

Every pool viewpoint v—including one occupied by a junk capture—must be covered, but the value of covering it with s is scaled by w_s, so a corrupted viewpoint is best covered by a nearby clean view rather than by its own low-w capture. A clean neighbor at distance d outbids a view's own junk capture whenever $d < \sigma\sqrt{2\ln(w_s/w_v)}$: the exchange rate between quality and coverage is set by the capture spacing itself, with no balancing hyperparameter (supplement). This "keep the viewpoint, swap the source" behavior is what lets QG-COVER degrade gracefully: as clean views grow scarce it trades a little coverage for quality, rather than selecting junk (FVS) or abandoning viewpoints (hard gate). F is *monotone* and *submodular*: each per-viewpoint term $\max_{s \in S} w_s \, k(v,s)$ can only grow as S expands, with diminishing returns, and both properties survive the sum over v. Greedy selection—at each step adding the view with the largest marginal gain—therefore attains at least a $(1 - 1/e) \approx 63\%$ fraction of the optimal coverage value [19]; the guarantee is on F, not on reconstruction PSNR. Algorithm 1 caches the $N \times N$ affinity matrix once and runs K greedy rounds, for $O(KN^2)$ selection time—negligible beside 3DGS training.

4. Experiments

Datasets and corruption. For controlled experiments we corrupt a deterministic 30% of the training frames of seven mip-NeRF-360 [1] scenes (room, counter, kitchen, bonsai, garden, bicycle, stump); synthesizing the corruption yields per-view ground-truth labels for the oracle gate and for evaluating ours as a detector. We model four corruption types: low-light (radiometric darkening plus read noise), sensor noise, defocus blur, and a heterogeneous *mixed* setting. The 30% rate was fixed before looking at any result, and it is close to what our gate later *measures* on the real captures (Sec. 4.2). For external validity we use real casual captures with *natural* quality variation and no synthetic corruption: 12 nerfbusters scenes and 14 DL3DV scenes—7 synthetic and 26 real scenes in total.

Trainers and protocol. All results in this paper are trained with LeGS [20], a 3DGS implementation whose learned density control replaces the original gradient-threshold heuristic; we keep its default settings and train for 7000 iterations. Selection is a pre-processing step: the trainer is the fixed downstream model every selector shares. Test views are every eighth frame by filename (on real captures additionally required to be of above-median no-reference quality, so PSNR is measured against a readable reference); the split is computed once, before any selector runs, and is byte-identical across methods, budgets and trainers. Synthetic held-out views are clean by construction: corruption touches only the training pool. We report PSNR in the main text and SSIM/LPIPS in the supplement.

Baselines. *Random* (uniform subset) and *FVS* (coverage-only); our methods are DETECT+COVER and QG-COVER. The supplement adds MaxDetail (the sharpness-maximizing, quality-only counterpart to FVS) and an oracle gate told which frames are corrupted, together with a DINOv2 content-coverage ablation. Selection cost is negligible: greedy selection on a typical pool ($N \leq 1000$) takes under 0.2 s against minutes per 3DGS fit.

Claims and falsification criteria. We pre-specified two claims with different evidentiary bars. *Claim 1 (matched-budget):* at equal budget, quality-gated selection beats coverage-only selection—falsified if QG-COVER does not beat FVS at K=40% on a majority of scenes, or if the margin does not survive a change of trainer or schedule. *Claim 2 (subset-beats-pool):* a well-chosen subset can outperform training on the entire pool—held to the higher bar of exceeding the full-pool result on a majority of scenes at a fixed schedule, and reported as *schedule- and domain-dependent* the moment it fails. Both are re-tested at the standard 30k schedule (supplement): Claim 1 survives; Claim 2 does not, though it holds on six of the seven synthetic scenes at 7k.

4.1. Main results

Budget accounting. The result that 40% of the pool can outperform 100% of it invites an obvious objection, so we measure the two competing effects directly: four models per scene on identical held-out views (Tab. 1): the uncorrupted pool in full (the true ceiling), a uniform 40% of it, the corrupted pool in full (the K=100% column of Tab. 2), and QG-COVER's 40% of the corrupted pool. On a *clean* pool

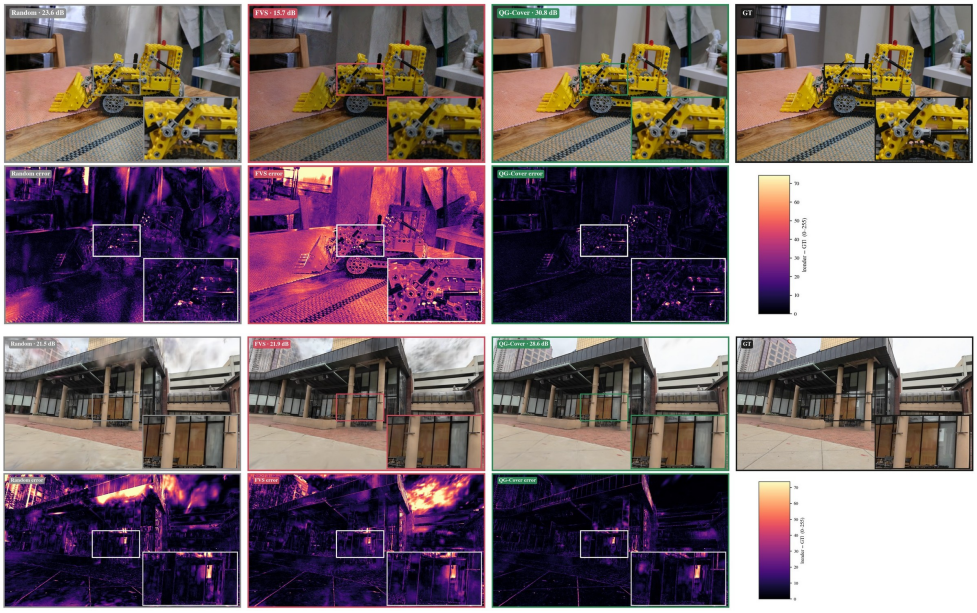


Figure 3. **Held-out novel views** on the *same* test frame, under controlled corruption (kitchen, mip-NeRF-360 with 30% synthetic junk; top) and on a genuine casual capture (DL3DV/0a4853, no synthetic corruption; bottom). Selections at the $K{=}40\%$ tier of Tab. 2. Render row: Random, FVS, QG-COVER (ours) and ground truth with per-image PSNR; error row: per-pixel $|\text{render} - \text{GT}|$ (magma, shared scale; darker is better). Zoom insets highlight the region where QG-COVER improves most over FVS. A representative gallery is in the supplement.

Table 1. **Budget accounting** (novel-view PSNR dB, ↑; identical held-out views throughout). Four models per scene isolate the two competing effects. Within the *clean* pool, dropping 60% of the frames costs only −0.99 dB on average (*coverage cost*), because casual capture is redundant; within the *corrupted* pool, spending that same 40% budget through our gate buys +1.08 dB (*gating gain*). That asymmetry, not the selection algorithm itself, is why a well-chosen 40% beats the whole corrupted pool. No selection reaches the clean-pool ceiling: our 40% subset still trails it by 1.57 dB.

	clean pool (no corruption)		*corrupted* pool (30% junk)		cost / gain	
scene	all frames ceiling	uniform 40% −60% frames	all frames $K{=}100\%$ col.	QG-COVER 40% ours	coverage cost clean 40% − clean all	gating gain QG-COVER − corrupt all
room	30.76	29.60	27.15	29.48	−1.17	+2.32
counter	27.54	26.93	25.76	25.77	−0.61	+0.00
kitchen	29.89	29.28	25.44	29.07	−0.61	+3.62
bonsai	30.06	30.02	26.84	28.66	−0.05	+1.82
garden	26.72	25.72	24.44	25.03	−1.00	+0.59
bicycle	23.50	22.63	22.34	22.82	−0.87	+0.48
stump	26.48	23.85	24.46	23.16	−2.63	−1.30
mean	**27.85**	26.86	25.21	26.28	−0.99	+1.08

more views *is* better—the uniform 40% loses on every scene (−0.99 dB mean)—but that penalty is small because casual capture is redundant, whereas keeping junk costs 2.64 dB against the clean ceiling; the +1.65 dB between the two is the headroom a perfect gate could claim, of which QG-COVER realises +1.08 dB. Crucially, our 40% subset does not reach the clean ceiling (1.57 dB short): view selection is damage control, not a way to manufacture information. The one reversal is instructive: stump has the smallest pool (109 frames) and the largest coverage penalty (−2.63 dB), and there the full pool wins.

Matched-budget comparison. The synthetic block of Tab. 2 compares the selectors at budgets set as a percentage of each scene's own pool. Both of our selectors recover a large, consistent margin at the tightest tier ($K{=}40\%$): DETECT+COVER +1.69 dB and QG-COVER +1.95 dB mean over FVS. The two then separate as the

Table 2. Novel-view PSNR (dB, ↑) versus selection budget K, expressed as a percentage of *each scene's own* candidate pool (pool sizes range 109–427 frames across scenes). Each dataset block lists every individual scene (all 33), followed by the *dataset mean*. Uniform Random (Rnd, unshaded baseline), FVS / DETECT+COVER / QG-COVER shown per budget tier; best / second-best of the three selectors per row shaded gold / grey. At K=100% every selector takes the entire pool and the methods converge by construction (Sec. 3.4) to one value (single unshaded column). SSIM/LPIPS companions and a fully expanded per-scene/per-metric table are in the supplement.

	K=40%				K=60%				K=80%				K=100%
Scene	Rnd	FVS	D+C	QG	Rnd	FVS	D+C	QG	Rnd	FVS	D+C	QG	(all methods)
Synthetic (mip-NeRF-360)													
room	27.10	26.78	28.29	29.48	27.66	27.63	27.52	29.23	27.32	28.16	28.10	28.74	27.15
counter	23.92	24.73	26.00	25.77	25.13	25.02	25.08	26.54	25.59	25.82	25.79	26.20	25.76
kitchen	24.15	25.26	29.28	29.07	26.25	26.25	25.48	29.06	25.99	26.69	26.24	27.46	25.44
bonsai	26.49	25.86	27.31	28.66	26.68	26.43	27.51	30.07	26.21	27.22	27.23	28.80	26.84
garden	22.97	23.83	25.86	25.03	24.33	25.09	26.24	25.80	24.60	24.85	25.08	25.85	24.44
bicycle	21.02	21.59	22.44	22.82	21.37	21.92	23.08	22.96	22.24	21.87	21.94	22.85	22.34
stump	22.34	22.25	22.94	23.16	23.57	22.96	24.86	23.85	23.70	23.59	23.56	24.15	24.46
dataset mean	24.00	24.33	26.02	26.28	25.00	25.04	25.68	26.79	25.09	25.46	25.42	26.29	25.21
nerfbusters													
nb-aloe	21.73	21.05	22.18	22.67	21.79	21.38	22.28	22.39	21.68	21.60	22.25	21.95	21.71
nb-art	22.80	23.26	23.23	23.72	22.86	23.01	23.06	23.33	23.00	22.98	22.99	23.08	23.17
nb-car	19.70	20.06	20.09	20.26	19.92	20.13	20.12	20.35	20.04	20.01	20.08	20.18	20.05
nb-century	23.30	23.16	23.34	23.85	23.18	22.44	23.49	23.74	23.35	23.14	23.85	23.53	23.23
nb-flowers	25.58	26.94	27.05	27.24	26.15	27.04	27.16	27.27	26.63	26.84	26.99	26.73	26.66
nb-garbage	21.95	22.02	23.39	23.01	22.48	22.04	22.07	23.14	22.10	22.10	22.23	22.38	22.16
nb-picnic	21.59	22.66	22.98	22.99	22.13	22.67	22.71	22.93	22.45	22.38	22.40	22.72	22.26
nb-pikachu	35.89	35.77	36.35	36.48	36.13	35.77	36.70	36.28	36.00	36.04	35.94	35.98	36.10
nb-pipe	21.84	22.09	22.87	23.17	22.16	21.98	21.99	22.81	22.05	22.12	22.18	22.42	22.34
nb-plant	31.35	32.04	32.64	32.96	31.26	31.73	32.57	32.45	31.60	31.89	31.88	32.09	31.71
nb-roses	30.85	31.58	31.86	32.19	31.34	31.55	31.77	31.70	31.50	31.58	31.60	31.57	31.56
nb-table	30.13	31.75	32.16	31.92	30.81	31.64	31.99	32.06	31.14	31.20	31.21	31.27	31.32
dataset mean	25.56	26.03	26.51	26.71	25.85	25.95	26.33	26.54	25.96	25.99	26.13	26.16	26.02
DL3DV													
dl3dv-032dee	28.64	29.50	31.14	29.89	29.58	29.74	29.89	30.36	29.67	29.82	29.50	30.17	29.48
dl3dv-0569e8	25.34	25.27	25.23	25.35	25.29	25.33	25.53	25.43	25.57	25.45	25.31	25.32	25.55
dl3dv-073f5a	19.92	20.65	20.95	20.90	20.43	20.70	20.96	20.97	20.54	20.79	20.81	20.77	20.63
dl3dv-07d9f9	34.21	34.87	34.90	35.16	34.36	34.87	34.96	34.85	34.56	34.30	34.17	34.69	34.67
dl3dv-085397	31.29	31.76	32.03	31.96	31.28	31.71	31.62	32.28	31.32	31.49	31.67	31.79	31.57
dl3dv-093ef3	27.61	28.10	28.49	28.40	27.54	27.96	28.09	28.38	27.70	28.10	28.09	28.27	28.03
dl3dv-0a1b7c	30.58	30.89	31.10	31.06	30.70	30.95	31.27	31.08	30.62	30.80	30.74	30.63	31.10
dl3dv-0a4853	27.23	29.46	30.14	30.50	29.17	28.58	29.51	30.07	29.03	29.66	28.91	30.03	29.26
dl3dv-0bfdd0	24.01	23.95	24.54	24.15	24.16	23.99	24.57	24.06	24.20	23.65	23.64	24.47	24.54
dl3dv-119fd5	25.07	25.61	25.81	25.65	25.11	25.74	25.88	25.92	25.49	25.49	25.56	25.77	25.67
dl3dv-15ff83	25.56	25.92	26.20	26.01	26.02	25.97	26.27	26.04	25.95	25.68	25.89	25.90	25.93
dl3dv-165f5a	23.11	23.47	23.82	23.63	23.42	23.46	23.71	23.65	23.23	23.43	23.37	23.73	23.34
dl3dv-1da888	28.32	29.29	30.01	29.97	29.18	29.57	30.02	29.82	29.21	29.44	29.40	29.51	29.14
dl3dv-1de58b	35.38	35.48	35.55	35.66	35.54	35.44	35.55	35.61	35.38	35.39	35.41	35.30	35.67
dataset mean	27.59	28.16	28.56	28.45	27.98	28.14	28.42	28.47	28.03	28.11	28.03	28.31	28.18

budget grows: QG-COVER leads the dataset mean at every partial tier, while the hard gate's edge decays to nothing (+1.69/+0.64/−0.04 dB at 40/60/80%)—once K approaches the number of clean frames, DETECT+COVER's fall-back must re-admit junk, whereas the soft objective keeps trading a little coverage for quality. Selecting well also beats selecting everything: at K=40% QG-COVER exceeds the full-pool result by the gating gain of Tab. 1 while training on 2.5× fewer views. Uniform random selection (Rnd in Tab. 2) confirms the diagnosis: FVS's mean edge over random never exceeds +0.6 dB on any dataset at any tier, it falls *below* random on 8 of 33 scenes at K=40% (13 of 33 at 60%), while QG-COVER beats random on *every* one of the 33 scenes at the tightest tier. At an aggressive *absolute* budget (K=40 frames, 15–37% of the seven synthetic pools) the diagnosis sharpens further: FVS falls below random indoors (room −0.80 dB).

4.2. Real scenes and qualitative results

The nerfbusters and DL3DV blocks of Tab. 2 report the real-data evaluation with no synthetic corruption; the effect is smaller but consistent: at the K=40% tier QG-COVER beats FVS by +0.68 dB on the 12 nerfbusters scenes and +0.29 dB on the 14 DL3DV scenes, and it is positive against FVS by strict sign on *all* 26 real scenes (mean +0.47 dB, Cohen's d=1.18). The margin shrinks as the

budget grows toward the forced tie at $K{=}100\%$, exactly as the mechanism predicts. These pools are corrupted in the same sense as our synthetic ones, only not by us: the gate flags 29%/33% of frames on nerfbusters/DL3DV (15–57% per scene; supplement), which is why the real curves also peak below $K{=}100\%$. The clean-pool control of Tab. 1 has no real-data analogue—no uncorrupted version of a real capture exists—so the corruption fraction there is measured, not known.

Qualitative results. Fig. 3 shows held-out renders on the same test frame for one controlled-corruption scene (kitchen, $17.9 \rightarrow 32.0$ dB) and one genuine casual capture (DL3DV/0a4853, $23.3 \rightarrow 26.9$ dB), both at the $K{=}40\%$ tier of Tab. 2. FVS trains on blurry anchors and smears texture; QG-COVER is visibly sharper with a cleaner error map, and the effect is the same with and without synthetic corruption. FVS repeatedly reconstructs a corrupted anchor that QG-COVER covers instead with a nearby clean capture; a representative gallery with error maps is in the supplement.

4.3. Discussion and limitations

The gain scales with the amount of *recoverable* corruption: scenes with a large, separable junk fraction see $+2$ dB or more, while near-clean scenes see little gain and can slightly favor FVS, consistent with our diagnosis. The procedure is nearly free: compute three image statistics, form the one-sided score (1), and either DETECT+COVER or run QG-COVER, the safer default when corruption is heavy or clean views are scarce. Our sweeps use one training-free pipeline with no learned component; cross-checks with the original 3DGS [6], FastGS [24] and UBS [14] are in the supplement. On real captures our evidence is the consistent direction rather than the magnitude, and we scope the paper to the quality-corrupted-pool *regime*. The test-view quality criterion only ensures readable held-out references on real captures (synthetic held-out views are unfiltered) and is fixed before selection for all methods, though the protocol does not evaluate rendering toward severely degraded targets. We reject rather than repair, without comparing against every corruption-specific repair trainer [10, 27, 36]; an equal-budget trimmed-loss test does not help on whole-frame corruption, and gating composes with such trainers. A learned utility predicting each view's downstream contribution is the one place a learned component might beat the cheap gate.

5. Conclusion

On casual, quality-varying capture, coverage-maximizing view selection for 3DGS is a liability: it over-selects the corrupted extremes and can fail to beat random. A training-free no-reference gate flips this, recovering $+1.69/+1.95$ dB at a 40%-of-pool budget on synthetic scenes—where the 40% subset also beats training on the whole pool—and winning on all 26 real scenes at that budget. Quality gating is the missing lever—coverage remains necessary, and a richer representation adds nothing once quality is controlled—and QG-COVER folds quality and coverage into a single submodular objective for graceful degradation under heavy corruption, as a trainer-agnostic front-end that costs three image statistics. We are deliberately narrow about the subset-beats-pool result: it is an accounting outcome, not a claim that data is dispensable—on a clean pool the full set wins, our 40% subset still trails clean-pool training by 1.57 dB, and the beaten baseline is only *this* pool's own 100% result, whose corrupted share is what makes it beatable (Tab. 1). In a redundant pool the marginal view is nearly free and the corrupted view expensive: the question at capture time is not "how many views" but "which *clean* views."

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